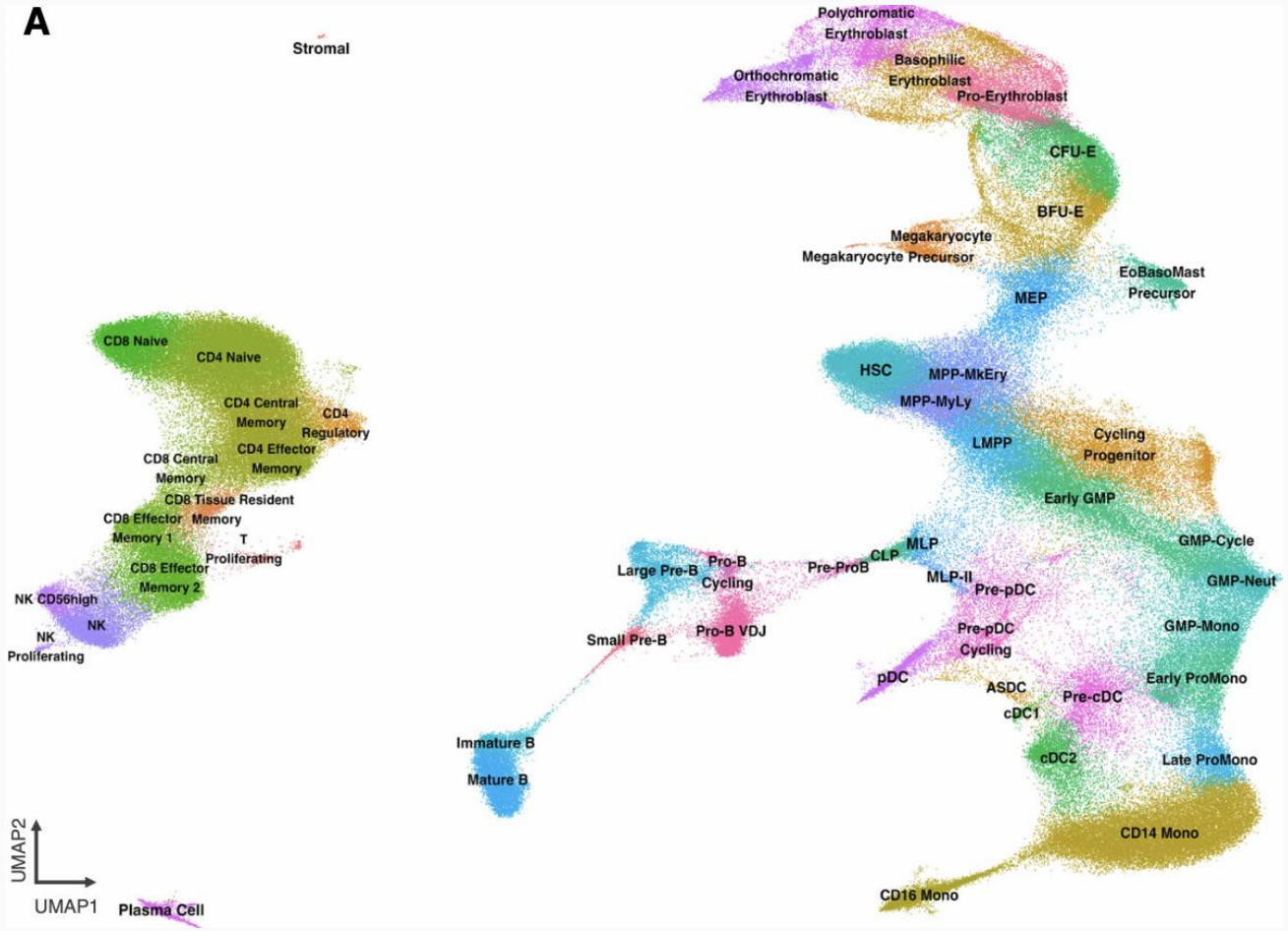
Interpreting Function Programs in Haematopoiesis by using SAE Autoencoder

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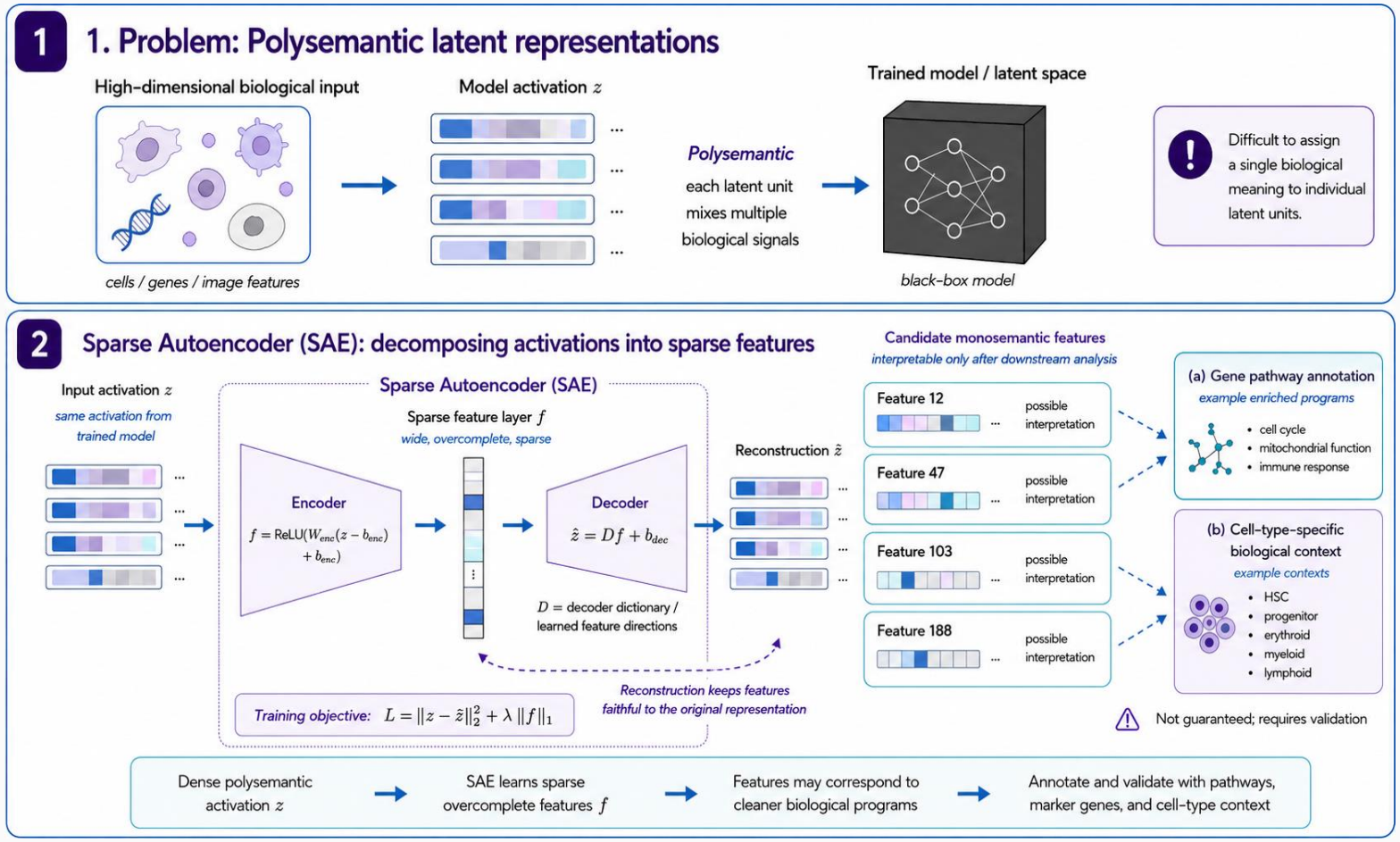
The Data: BoneMarrowMap Haematopoietic Atlas



263,159 cells | 161,604 for SAE training | BoneMarrowMap, Zeng et al. 2025

- 34 annotated cell types spanning the full haematopoietic hierarchy
- **Method: cNMF**
  (Consensus Non-negative Matrix Factorization)
- **Subset:** Unsorted + Healthy Reference donors
- **scVI (Single-cell Variational Inference) Training:**
  number of latent=30,
  number of top genes = 4000,
  number of batch key = Study
- **Latent representation:**
  128-dim encoder intermediate layer extracted for SAE input

Polysemantic Representations: The Challenge



- Neural network latent spaces encode multiple unrelated concepts per dimension (superposition)
- Sparse Autoencoders (SAEs) decompose polysemantic representations into 209 monosemantic features
- Each active feature corresponds to a single interpretable biological programme
- Sparsity constraint: L0 = 53.5 active features per cell on average
- Reconstruction quality: frac variance explained = 0.9955

SAE extracts monosemantic features from polysemantic scVI latent vectors.

Zero-shot Annotation: Prompt Design

System Prompt

You are a computational biology expert specializing in Haematopoiesis and single-cell transcriptomics. You will be shown a gene signature extracted by a Sparse Autoencoder (SAE) trained on scVI latent representations of the BoneMarrowMap atlas (~160k human bone marrow cells). Your task is to identify the biological process or functional programme this feature represents.

Input Categories

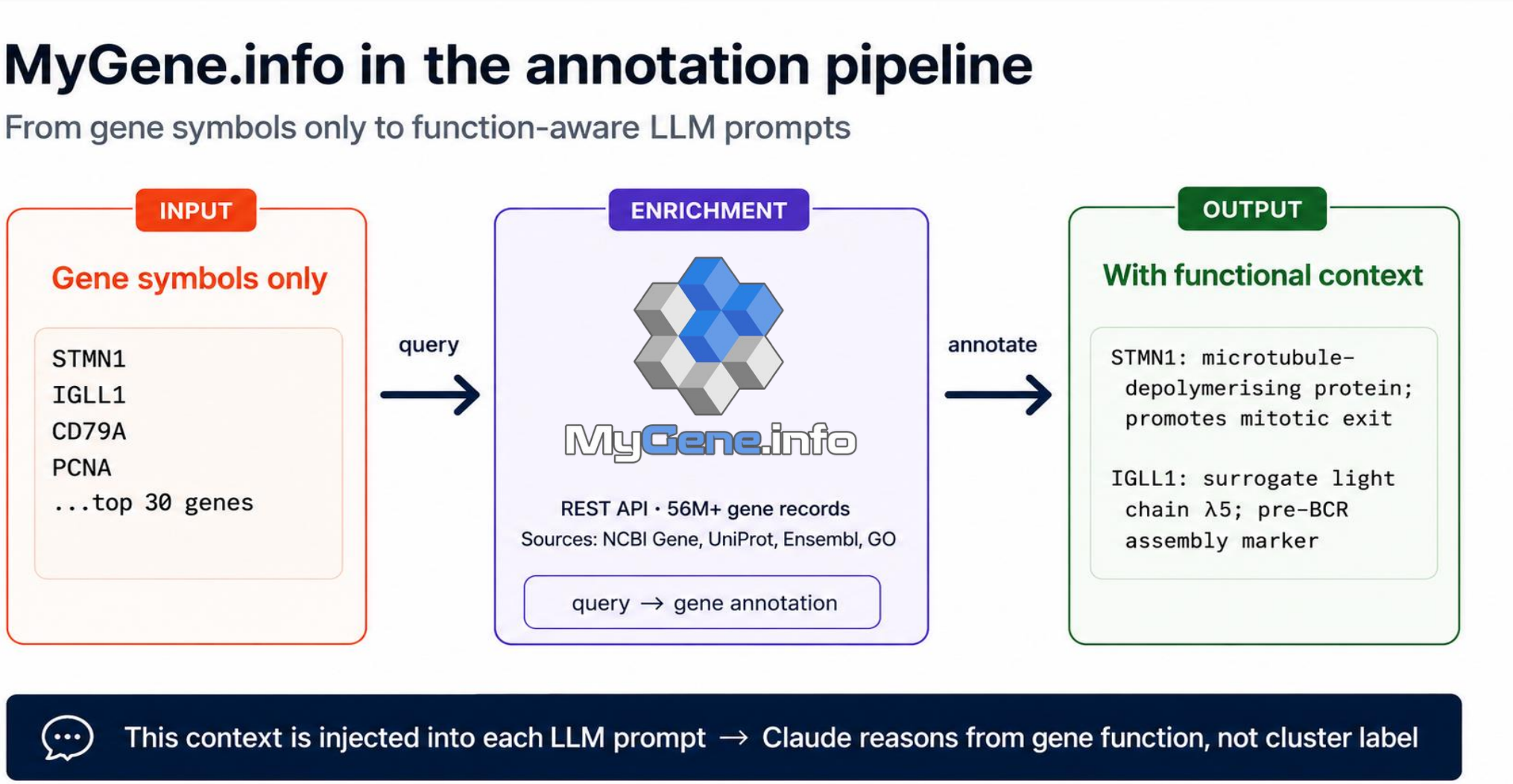
- Feature ID + ANOVA rank
- Top 30 genes (t-statistic ranking): e.g. STMN1 / PCNA / IGLL1 / CD79A
- Cell-type composition of top-200 activating cells (%) e.g. Large Pre-B: 70%, Pro-B Cycling: 29.5%

- Activation fold-enrichment: top-200 mean / global mean
- NO gene function context: Gene symbols only: LLM must infer from memory

Output Schema

```
{
  "programme_name":          "≤ 6 words",
  "functional_description":  "2–3 sentence biological explanation",
  "key_evidence":            "up to 5 genes + rationale per gene",
  "confidence":              "HIGH / MEDIUM / LOW",
  "suggested_validation":    "recommended follow-up experiment"
}
```

What is MyGene.info?



RAG Enhanced LLM Annotation

MyGene.info retrieval injects gene function context so Claude can distinguish sub-programmes

202/209
Updated by RAG

6/10
Top Features Closer to Human

F#332
Critical Misclassification Fixed

CASE STUDY: FEATURE #332 LINEAGE MISCLASSIFICATION CORRECTED BY RAG

BEFORE: Zero-shot LLM (no gene context)

Programme name:
ER stress and secretory protein processing

Confidence: MEDIUM

Reasoning:
Gene list matched ER stress signatures.
LLM followed cluster label, not actual gene functions.

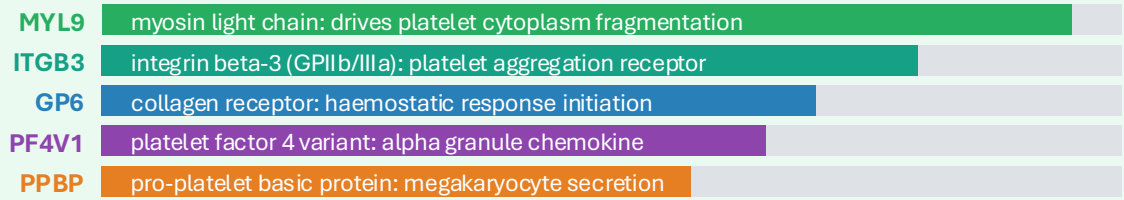
Root cause:
Cluster 1 label = ER folding. LLM adopted the label without inspecting what MYL9 or ITGB3 actually do.

AFTER: RAG with MyGene.info context

Programme name:
Megakaryocyte cytoskeletal and platelet assembly programme

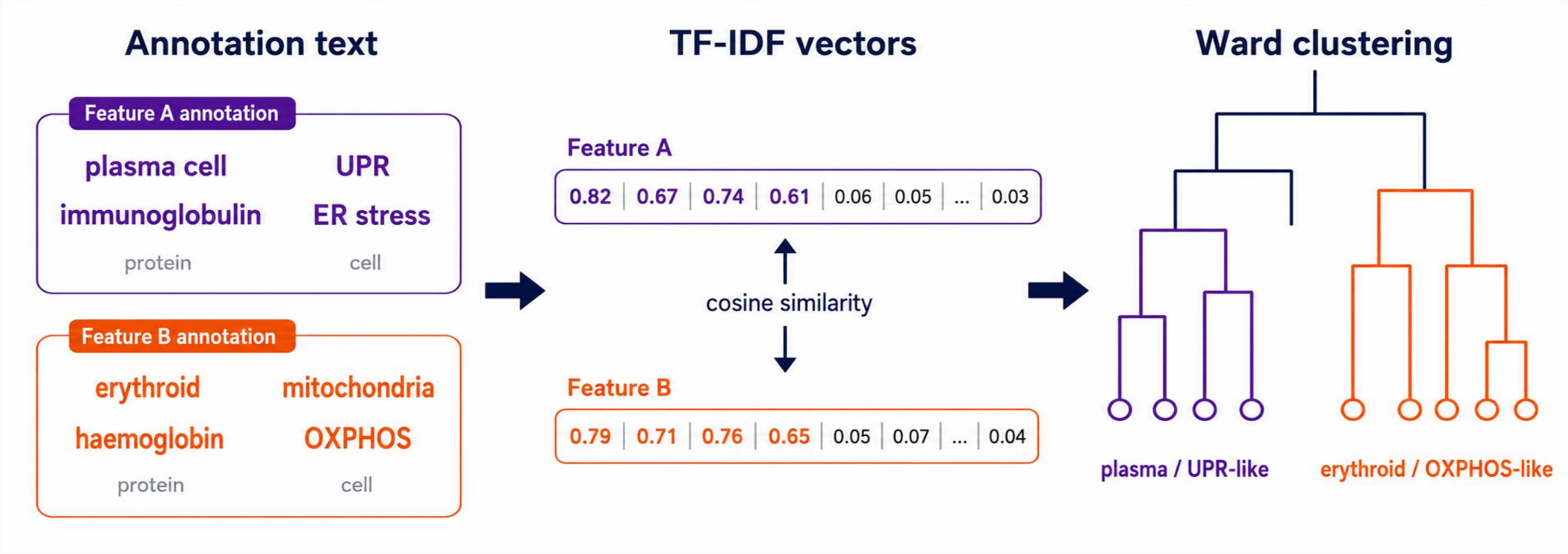
Confidence: HIGH

Key evidence genes:



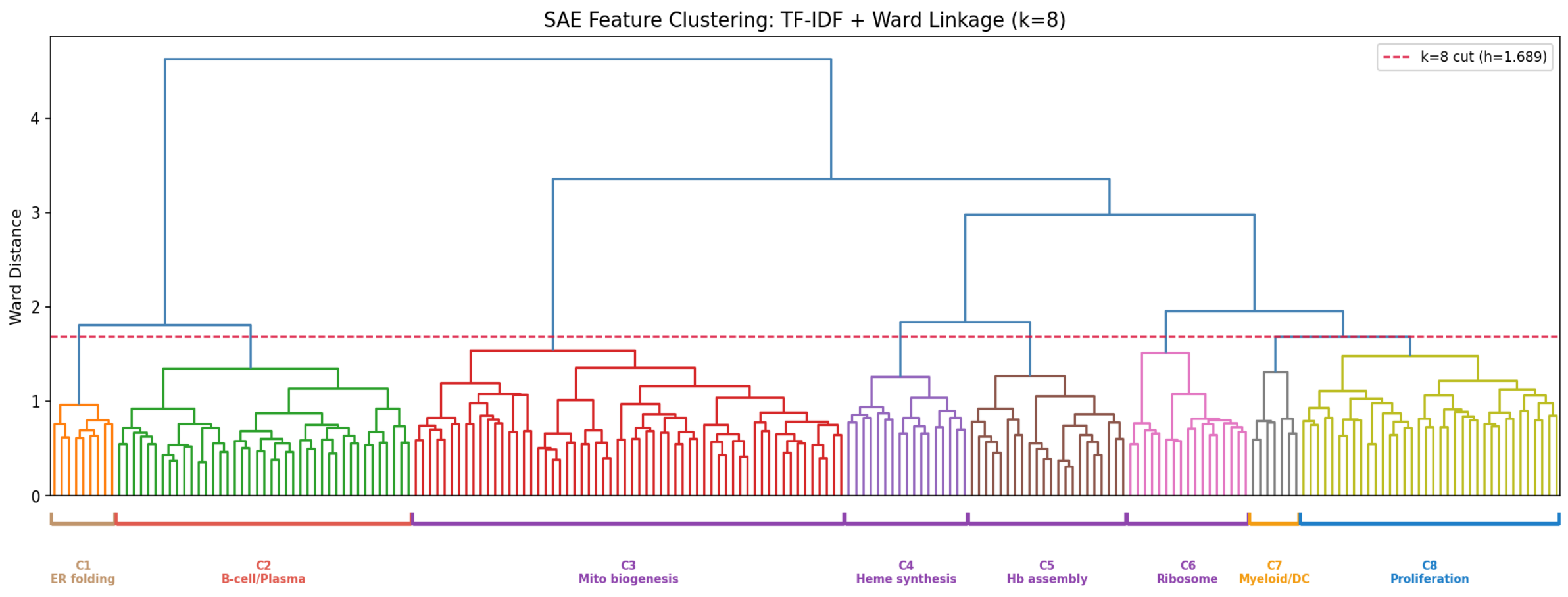
CONCLUSION: RAG forces the LLM to reason from gene function, not cluster label, correcting a systematic misclassification zero-shot LLM cannot detect.

Clustering Method: TF-IDF



209 annotations → TF-IDF vectorisation (sklearn, bigrams) → Ward linkage → k=8 clusters

Functional Program Tree: 209 Active Features, 8 Clusters



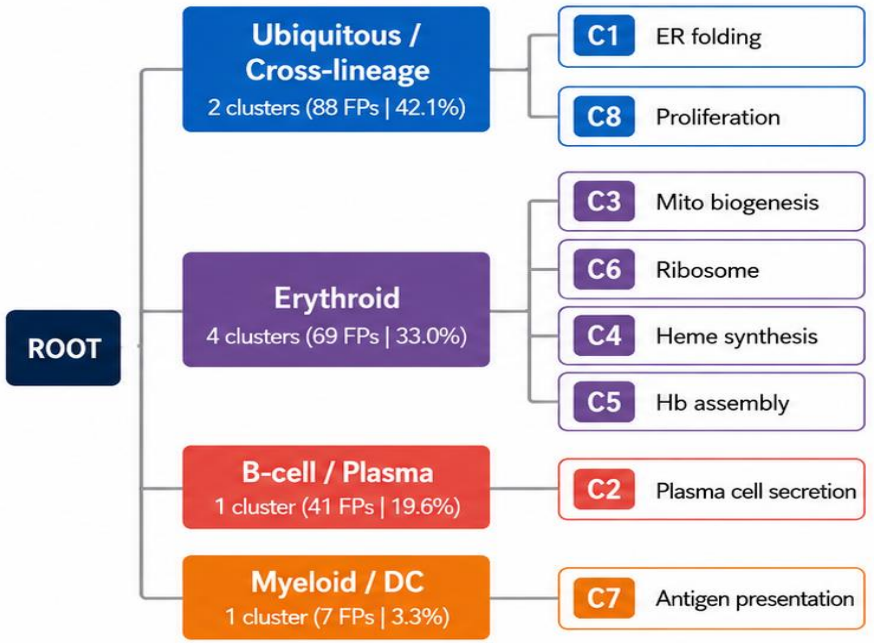
8 clusters via TF-IDF + Ward. Erythroid: 4 stages (mito, ribosome, haem, Hb).

Functional Program Tree:
209 Active Features, 8 Clusters

SAE Feature Taxonomy from TF-IDF + Ward Linkage (k = 8)

Readable summary of dendrogram-derived functional clusters

Derived from dendrogram clustering



Cluster	Higher-order module	Representative functional programme / interpretation	Example markers or themes
C1	Ubiquitous / Cross-lineage	ER protein folding and chaperone	ER folding, oxidative folding, chaperones
C8	Ubiquitous / Cross-lineage	Progenitor proliferation	E2F, MYC, cell cycle, proliferation
C3	Erythroid	Mitochondrial biogenesis (OXPHOS)	mitochondria, OXPHOS, bioenergetics
C6	Erythroid	Ribosome biogenesis	RPL, RPS, translational machinery
C4	Erythroid	Heme synthesis	ALAS2, FECH, porphyrin / heme pathway
C5	Erythroid	Hemoglobin assembly	HBA1, HBB, globin programme
C2	B-cell / Plasma	Plasma cell Ig secretion + UPR/ERAD	immunoglobulin secretion, UPR, ERAD
C7	Myeloid / DC	MHC-II antigen presentation	HLA-DR, antigen presentation

Clusters are summarized from the k = 8 TF-IDF + Ward linkage solution and then biologically annotated by dominant functional programmes.
Totals: 8 clusters, 209 active features (100%).

8 clusters via TF-IDF + Ward. Erythroid: 4 stages (mito, ribosome, haem, Hb).

SAE vs cNMF on Same Dataset
What SAE Uniquely Enables

SAE VS EXISTING METHODS: COMPARATIVE ASSESSMENT

Dimension	cNMF (Zeng 2025)	scVI (baseline)	SAE (this work)
Programme resolution	48 coarse programmes	30 polysemantic dims	**209 monosemantic features**
Erythroid resolution	1 entry (single stage)	Embedded in dim 12+	**4 sequential stages**
Interpretability method	Human-curated gene weights	Manual marker correlation	**Automated LLM annotation**
Hierarchy	None	None	**FP-Tree (8 clusters)**
External validation	Marker gene overlap	UMAP visualisation	**GSEA/ORA 10/10**
Scalability	Requires re-training	Requires re-training	**Annotation pipeline: 30 min**

Next Steps

1. **Contrastive gene ranking for Cluster 2 plasma cell sub-programmes**
   Global t-test produces identical gene lists for all 41 features (Jaccard 0.592). Within-cluster contrastive t-test (Feature A top-200 vs Feature B top-200) to disambiguate UPR / ERAD / Ig-secretion sub-programmes.

2. **Quantitative SAE vs cNMF benchmarking:**
   Jaccard and cosine heatmap: 209 SAE features vs 48 cNMF programmes (Zeng et al. 2025) on the same BoneMarrowMap dataset. Quantify how many cNMF programmes each SAE cluster recovers.

3. **Full haematopoietic lineage coverage:**
   Current B-cell ANOVA captures 19.6% of active features. Extend ANOVA to T-cell, NK, myeloid, and megakaryocyte lineages to complete the FP-Tree hierarchy.

4. **RAG enrichment and TF regulon validation:**
   Add UniProt, CellMarker 2.0, STRING as additional retrieval sources. Validate FP-Tree clusters against GATA-1, PU.1, and RUNX1 regulons as biological ground truth.

Thank you for Listening!



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